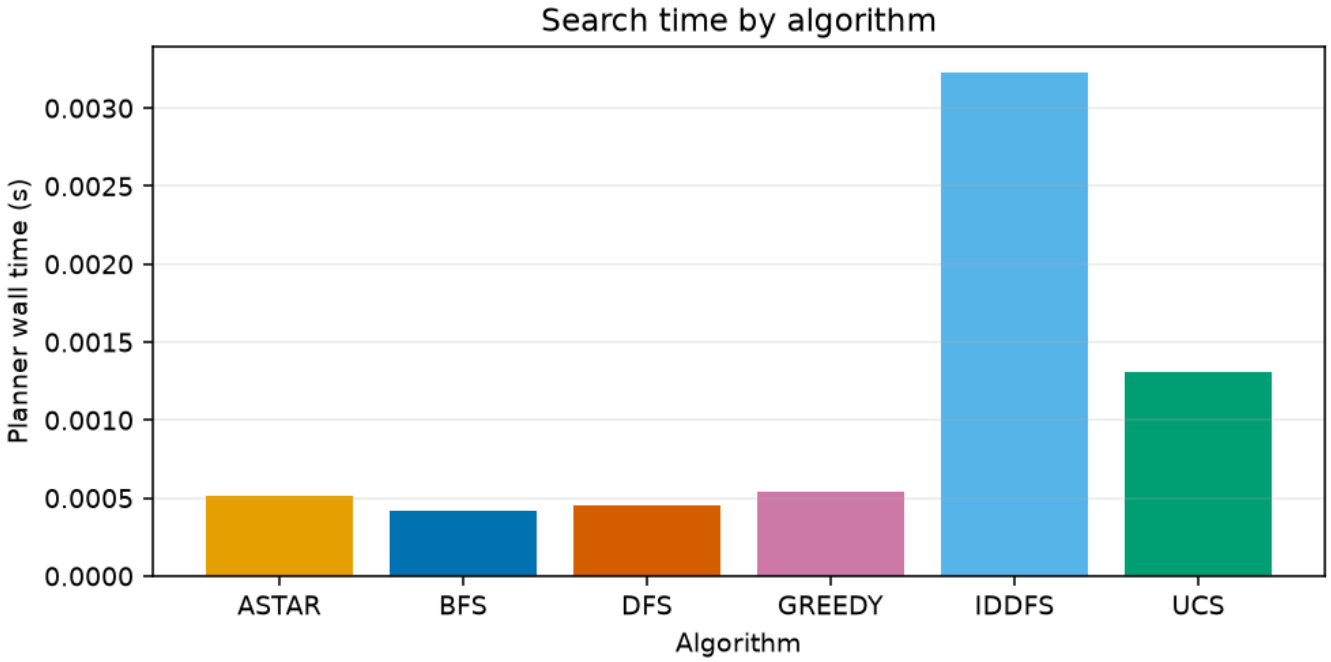
Point To Point comparison

Runs share map, robot, policy, neighbor order, cost model, heuristic, and mission mode.

Algorithm	Expanded	Search (s)	Steps	Cost	Energy	Coverage %	Stop
ASTAR	10	0.0005136999534443021	10	10.0	0.6999999999999318	N/A	mission_complete
BFS	30	0.00041630002669990063	10	10.0	0.6999999999999318	N/A	mission_complete
DFS	10	0.00045469996985048056	10	10.0	0.6999999999999318	N/A	mission_complete
GREEDY	10	0.0005377000197768211	10	10.0	0.6999999999999318	N/A	mission_complete
IDDFS	131	0.003230600035749376	10	10.0	0.6999999999999318	N/A	mission_complete
UCS	30	0.0013119999784976244	10	10.0	0.6999999999999318	N/A	mission_complete



Assumptions and limitations

Map grid: 6 × 6; resolution 1.0 m/cell. Movement: four-neighbor-cardinal; neighbor order: ['up', 'right', 'down', 'left']; heuristic: manhattan; cost model: {'transition': 'unit'}; tie-breaking: recorded algorithm default; coverage policy: None.

A single scenario does not support a generalized superiority claim. DFS depends on neighbor order. Greedy search is not guaranteed to return an optimal path. IDDFS may repeat expansions across depth limits. Simulated battery values are abstract model outputs, not real appliance measurements.